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Estimation of gillnet and hook selectivity using log-linear models

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The log-linear modelling capabilities of existing statistical software permit a generalization of Holt's indirect estimation method by allowing several different selection curve models to be fitted to catch data from an arbitrary number of mesh sizes. This also facilitates the use of formal inferential procedures such as model selection, assessment of relative fishing powers, estimation of standard errors, and permits inclusion of knowledge regarding the shape of the population length distribution. This methodology is equally applicable to estimation of hook selectivity.

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Key words: count data, gillnet selectivity, hook selectivity, log-linear models, selection curves, SELECT method, maximum likelihood.

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Introduction

Knowledge of the size-selectivity of commercial fishing gears is crucial to management of a fishery for purposes of maximizing yield and protecting juvenile fish (Gulland, 1983; Wileman *et al.*, 1996). Moreover, fishing gears may be used as research tools for monitoring the length distribution of the stock by using the size-selectivity of the gears to adjust the length distribution of the catches. Gillnets are widely used for this purpose (Hamley, 1975).

Gillnet selectivity experiments are typically implemented by the simultaneous fishing of several gillnets of differing mesh sizes. If the length distribution of the fished population is ''known'' then selectivity can be estimated directly. Good knowledge about the population length distribution is rare and in practice one might consider an experiment that used only the recaptures of a tagged sub-population of Fish (e.g. Hamley and Regier, 1973; Myers and Hoenig, in press). More commonly, direct estimation is not feasible, whence indirect estimates of gillnet selectivity are obtained by comparing the observed catch frequencies across the various meshes fished.

Methods for calculating indirect estimates of gillnet selectivity from comparative catch data have been provided by Holt (1963), Regier and Robson (1966), Hamley (1975), Kirkwood and Walker (1986), Boy and Crivelli (1988), Helser *et al.* (1991, 1994), Henderson and Wong (1991), Millar (1992), and others (see Holst and Moth-Poulsen (1995) for a brief description of several of these methods, including their application to a common dataset). The methods of Kirkwood and Walker (1986) and Millar (1992) utilize the same underlying statistical model, and it is this model that is developed further here. The other approaches do use statistical tools (e.g. linear or non-linear regression) to varying degrees, but they are used outside of the context of a statistical model appropriate to gillnet catch data. For example, the linear regression approach of Holt (1963) does not model the data as counts and must be applied multiple times because it can only be applied to pairs of gillnets. Hence, the statistical properties of the resulting selectivity estimates are largely unknown. The recent studies of Helser *et al.* (1991, 1994) and Henderson and Wong (1991) do not model selectivity but instead follow an historical approach (Hamley, 1975) of referring to the catch length distribution as the selection curve. Such approaches do not estimate retention probabilities and hence are not of interest to this present study.

In this paper we present a general statistical model that is appropriate for the estimation of gillnet selection curves (i.e. retention probabilities) from comparative gillnet catch data. In many cases the model is log-linear. Indeed, it was the log-linear reduction that was utilized by Holt (1963) to estimate normal shaped selection

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curves using catch data from pairs of similar sized mesh gillnets. Here we refine the approach and make it appropriate to count data from an arbitrary number of mesh sizes, to other selection curve shapes, and explicitly consider the issue of relative fishing power of the meshes. Several such models are fitted to gillnet catch data of Fraser River sockeye salmon (Holt, 1963) using standard statistical software.

Materials and methods

For a given length class, L, the numbers of fish, y_{Lj}, that encounter gillnet j are assumed to be observations of independent Poisson random variables,

$$Y_{Lj} \sim Po(p_j \lambda_L)$$

where the expected count, $p_j\lambda_L$, is the product of the abundance of length class L fish, λ_L, and the relative fishing intensity of gillnet j, p_j. Relative fishing intensity (Millar, 1992) of a gillnet is a combined measure of fishing effort and fishing power.

If a count has a Poisson distribution then its variance is equal to its expected value (McCullagh and Nelder, 1989). However, count data from biological experiments often exhibit variances well in excess of this. This phenomenon is known as overdispersion, and in this present application it could be caused by grouping behaviour such as the schooling of fish. To allow for overdispersion the data could alternatively be modelled as observations of independent negative binomial random variables. However, general practice indicates that the effect of overdispersion on estimated parameters is negligible and inferential procedures can be suitably amended (McCullagh and Nelder, 1989, p. 200).

Denote the retention probability of length L fish in gillnet j by $r_j(L)$. The number of length L fish caught in gillnet j, N_{Lj}, is then distributed (Feller, 1968) as

$$N_{Lj} \sim Po(p_j \lambda_L r_j(L)). \quad (1)$$

Without loss of generality it can be assumed that the selection curves $r_j(\cdot)$ for each gillnet have unit height because any differences in fishing powers of the gillnets is modelled through the relative fishing intensities p_j.

So far, the model has been presented in full generality. In practice the researcher will be required (Millar, 1995) to make assumptions and/or inferences about the form of p_j, λ_L and $r_j(\cdot)$. Options to be considered include:

1. If the gillnets are fished with equal effort then should the relative fishing intensities, p_j, be assumed equal, proportional to mesh size, some other function of mesh size, or should no assumptions be made? (For a detailed discussion see Hamley, 1975).
2. Is it reasonable to postulate a form for the population length distribution (through specification of λ_L)? For example, is the population length distribution normally distributed, lognormally distributed, bimodal, or should no assumptions on λ_L be made?
3. Is the selection curve normal, gamma, lognormal shaped, or perhaps bimodal? Does the principle of geometric similarity (length of maximum retention and spread of selection curve both proportional to mesh size, Baranov, 1948) apply?

In all cases, one can use the Poisson distribution of the N_{Lj} in Equation (1) to apply maximum likelihood for purposes of statistical estimation and inference. Many of these maximum likelihood fits can be programmed with moderate effort if a reliable optimizing algorithm is available. However, a thorough statistical analysis will require programming that implements a number of the different options listed above. It will be necessary to calculate model deviances (likelihood ratio goodness of fit statistics) from each fit, and the ability to plot residuals is highly desirable.

Fortunately, for many choices the expected value, $v_{Lj} = p_j \lambda_L r_j(L)$, of the catch of length L fish in gillnet j can be expressed in log-linear form. That is, $\log(v_{Lj})$ is a linear combination of the form

$$\log(v_{Lj}) = \log(p_j) + \log(\lambda_L) + \log(r_j(L)) = \sum_i \beta_i \cdot f_i(j, L),$$

where $f_i(j,L)$ denotes a term that is a function of only j and/or L. In such cases the maximum likelihood model is a log-linear model and hence can be easily fitted using existing statistical software. Model deviances and residuals are routinely provided by such software.

By way of example, if (i) the gillnets have equal fishing intensity, (ii) the form of the population length distribution is not assumed, and (iii) the selection curves are assumed to be normal shaped and to observe geometric similarity (i.e. mean and spread proportional to mesh size) then the selection curve parameters are estimated from a fit of the log-linear model (appendix A)

$$\log(v_{Lj}) = \text{factor}(L) + \beta_1 \cdot x_{Lj} + \beta_2 \cdot x^2_{Lj} \quad (2)$$

where $x_{Lj} = L/m_j$ and m_j is the mesh size of gillnet j. Here, factor(L) denotes that length class is fitted as a factor in the model. The maximum likelihood fit of this model is included in most general purpose statisical software packages and in Appendix B we provide the implementation of this model in SAS (1988) and Splus (Becker *et al.*, 1988; Chambers and Hastie, 1992).

Table 1. Normal (fixed spread), normal (proportional spread), gamma, and log-normal selection curves. All are of unit height because relative fishing intensity is modeled separately. The right hand column gives the last two terms in the log-linear model. See Equations (2) and (4) for the normal (proportional spread) example.

Model	Selection curve	$[\beta_1]\{f_1(j,l)\}+[\beta_2]\{f_2(j,l)\}$
Normal: fixed spread	$\exp\left(-\frac{(L-k\cdot m_j)^2}{2\sigma^2}\right)$	$\left[\frac{k}{\sigma^2}\right]\cdot\{L\cdot m_j\}+\left[-\frac{k^2}{2\sigma^2}\right]\cdot\{m_j^2\}$
Normal: spread $\propto m_j$	$\exp\left(-\frac{(L-k_1\cdot m_j)^2}{2k_2\cdot m_j^2}\right)$	$\left[\frac{k_1}{k_2}\right]\cdot\left\{\frac{L}{m_j}\right\}+\left[-\frac{1}{2k_2}\right]\cdot\left\{\left(\frac{L}{m_j}\right)^2\right\}$
Gamma spread $\propto m_j$	$\left(\frac{L}{(\alpha-1)\cdot k\cdot m_j}\right)^{\alpha-1}\exp\left(\alpha-1-\frac{L}{k\cdot m_j}\right)$	$[\alpha-1]\cdot\left\{\log\left(\frac{L}{m_j}\right)\right\}+\left[-\frac{1}{k}\right]\cdot\left\{\frac{L}{m_j}\right\}$
Lognormal spread $\propto m_j$	$\frac{1}{L}\exp\left(\mu_1+\log\left(\frac{m_j}{m_1}\right)-\frac{\sigma^2}{2}-\frac{\left(\log(L)-\mu_1-\log\left(\frac{m_j}{m_1}\right)\right)^2}{2\sigma^2}\right)$	$\left[\frac{1}{\sigma^2}\right]\cdot\left\{\log(L)\cdot\log\left(\frac{m_j}{m_1}\right)-\frac{1}{2}\log^2\left(\frac{m_j}{m_1}\right)\right\}+\left[1-\frac{\mu_1}{\sigma^2}\right]\cdot\left\{\log\left(\frac{m_j}{m_1}\right)\right\}$

Table 2. Fraser River sockeye catch data from Holt (1963).

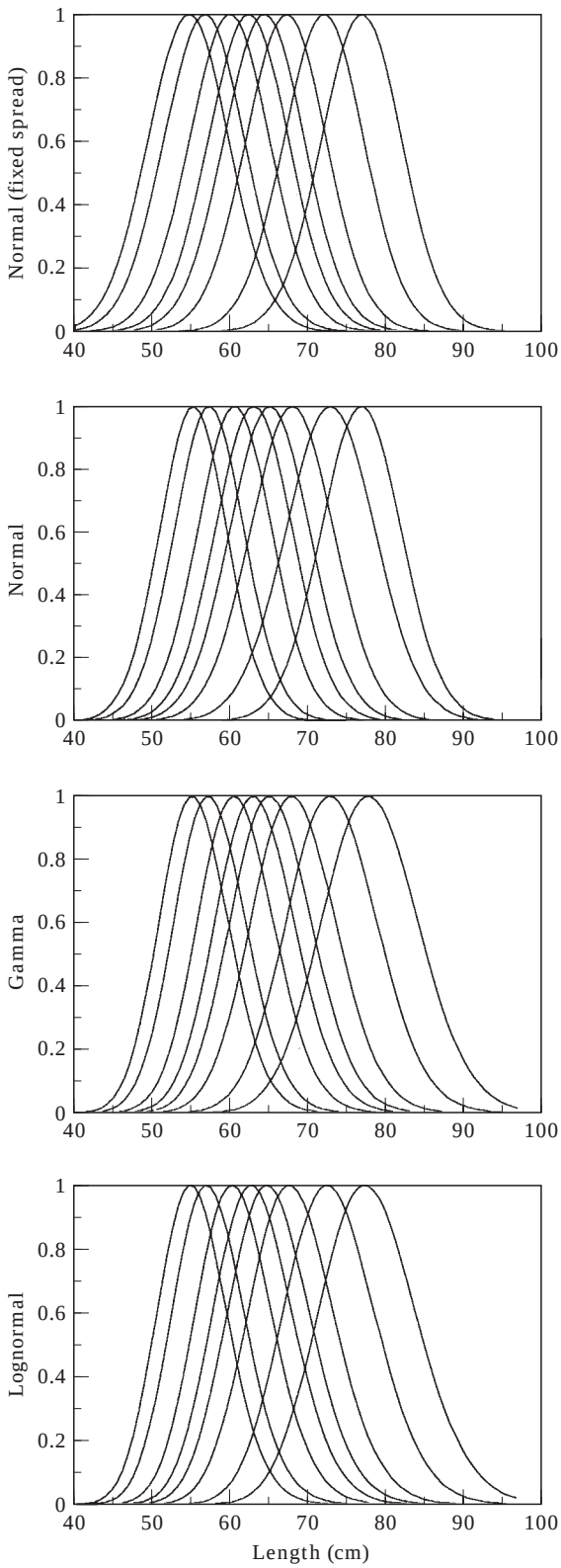
Fork length (cm)	Meshsize (cm) 13.5	14.0	14.8	15.4	15.9	16.6	17.8	19.0
52.5	52	11	1	1	0	0	0	0
54.5	102	91	16	4	4	2	0	3
56.5	295	232	131	61	17	13	3	1
58.5	309	318	362	243	95	26	4	3
60.5	118	173	326	342	199	100	10	11
62.5	79	87	191	239	202	201	39	15
64.5	27	48	111	143	133	185	72	25
66.5	14	17	44	51	52	122	74	41
68.5	8	6	14	23	25	59	65	76
70.5	7	3	8	14	15	16	34	33
72.5	0	3	1	2	5	4	6	15

Example

Log-linear modelling was used to fit normal (fixed spread), normal (proportional spread), gamma and lognormal selection curves (Table 1) to Fraser River sockeye salmon data (Table 2). The latter three selection curves observe geometrical similarity. Each selection curve was fitted twice, first under the assumption of equal fishing power of the gillnets and then again assuming fishing power to be proportional to mesh size (Table 3, Fig. 1). Each mesh size was fished with equal effort and hence fishing power is the same as fishing intensity in this example.

For both normal selection curve models (fixed and proportional spread) the model deviance was lower (indicating a better fit) under the assumption of equal fishing power of the different sized meshes (Table 3). The model deviances from the gamma and lognormal selection curve models were not influenced by the fishing power assumption. Overall, the lognormal selection curve provided the best fit. However, the model deviance of 704.3 on 75 d.f. does indicate overdispersion and/or severe lack of fit. A plot of deviance residuals (McCullagh and Nelder, 1989) shows that there is indeed lack of fit (Fig. 2). These deviance residuals are precisely the same for both the equal fishing intensity and proportional fishing intensity fits of the lognormal shaped selection curve.

The residual plot reveals some curious features of these data. It appears that the fishing powers of the smallest (13.5 cm) and largest (19.0 cm) meshes are greater than modelled because of the predominance of positive residuals. Also, for these two meshes the biggest (positive) residuals are for catches of the smaller and larger length classes, suggesting that they caught substantially higher than expected catches of these sizes of



salmon. The converse appears to apply for the other six mesh sizes, with less of the smaller and larger length classes caught than expected.

Discussion

The approach of Holt cannot use the full set of catch data (Table 2) because it calculates ratios of catches across pairs of gillnets and to avoid highly variable (or non-existent) ratios it is necessary to exclude counts that are not considered sufficiently large (Holt, 1963, p. 108). In contrast, the analysis presented above does use the full data because the likelihood obtained from Equation (1) is appropriate to count data and hence is valid for small or zero catches.

With no prior assumptions on the population length distribution the length class variable is fitted as a factor and the model is then equivalent (McCullagh and Nelder, 1989, p. 209) to the SELECT (Share Each Length's Catch Total) model of selectivity (Millar, 1992; Millar and Walsh, 1992). When a population length distribution is assumed the general model may reduce to the log-linear form in some circumstances. For example, assuming a normal population length distribution and normal shaped selection with proportional spread, the log-linear form is attained (Appendix A). For the Fraser River data this resulted in a population length distribution with an estimated mean length of 60.5 cm and standard deviation of 4.5 cm. The model deviance was 987.3 on 83 d.f. In comparison to the model with no assumed length distribution this is an increase in deviance of 214.5 on a difference of 8 d.f. Notwithstanding the lack of fit noted in the previous section, this suggests that the assumed normal shaped length distribution is not appropriate.

In many populations the mixture of different year-classes of recruited fish will result in a bimodal or multimodal length distribution curve. The genera l model does not reduce to the log-linear form in these cases. However, some statistical software have implemented recently developed smoothing techniques (Hastie and Tibshirani, 1990; Chambers and Hastie, 1992) which enable the user to specify that the population length distribution be a ''smooth function'' of length without the need to specify a parametric form (Appendix B).

The model does not reduce to the log-linear form if the selection curve is of the bimodal form that results from a combination of two selection mechansims

Figure 1. The selection curves (Table 3) fitted to the Fraser River sockeye salmon data. These curves assume equal fishing power of the gillnets – the curves obtained assuming fishing power proportional to mesh size are virtually indistinguishable from these.

Table 3. Log-linear fits to the data of Table 2. The model deviance is the likelihood ratio goodness of fit statistic and it has 75 degrees of freedom for each of the models shown.

Model	Equal fishing powers		Fishing power α mesh-size	
	Parameters	Model deviance	Parameters	Model deviance
Normal:				
fixed spread	$(k, \sigma) = (4.044, 5.148)$	862.9	$(k, \sigma) = (4.072, 5.180)$	883.6
spread α m_j	$(k_1, k_2) = (4.095, 0.1035)$	772.8	$(k_1, k_2) = (4.120, 0.1029)$	773.2
Gamma:				
spread α m_j	$(\alpha, k) = (158.3, 0.02592)$	719.3	$(\alpha, k) = (157.3, 0.02592)$	719.3
Lognormal:				
spread α m_j	$(\mu_1, \sigma) = (4.012, 0.0803)$	704.3	$(\mu_1, \sigma) = (4.006, 0.0803)$	704.3

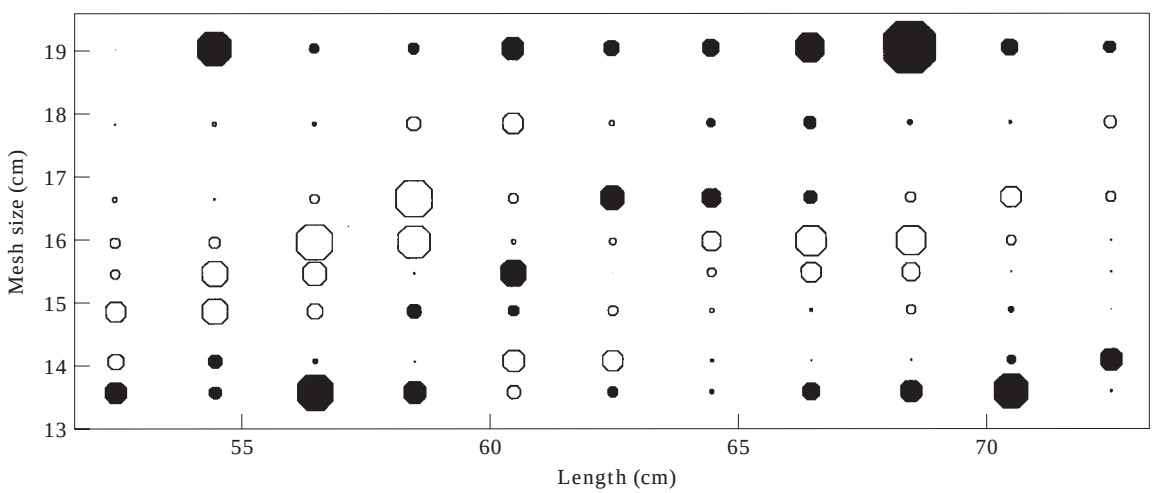


Figure 2. Deviance residual plot for the lognormal selection curve fits (Table 3) to Fraser River sockeye salmon data. Solid and open octagons represent positive and negative residuals respectively. The area of the octagon is proportional to the square of the residual.

(Hamley, 1975) such as wedging and tangling. Also, the fixed spread gamma selection curves specified by Kirkwood and Walker (1986) cannot be put in log-linear form because the scale parameter β (see Equation 7 on p. 693 of Kirkwood and Walker, 1986) cannot be expressed in the form kf(m). Here k is a parameter to be estimated and f(m) denotes a function of mesh size, m. In these cases the researcher will need to develop customized software to fit the curves using the likelihood derived from Equation (1).

Direct experiments of Hamley and Regier (1973) and Borgstrøm (1989) provide strong evidence of increased fishing power with mesh size in their studies (see also Hamley, 1975). If the relationship between mesh size and fishing power is assumed known then it can be included in the log-linear model (Appendix A) through the use of offsets (McCullagh and Nelder, 1989). Otherwise, it can be estimated within the general framework of the model presented here.

This present work is not intended to be a complete analysis of the Fraser River data, but rather to show how indirect estimation from gillnet catch data should proceed and the possibilities for exploration that already exist using existing statistical software. Future research will explore the possibilities and limitations of this general methodology and its ability to make inference associated with gillnet selectivity, including questions about relative fishing powers, population length distributions and shapes of selection curves. One known limitation of indirect estimation is that two different models may give precisely the same fit (Millar, 1995). Therefore, it can never be established that a particular selection curve is the ''right one''. The inability to distinguish models occurred in the current analysis – fits of gamma or lognormal selection curves do not change when the fishing powers of the gillnets are assumed proportional to mesh size because the offset is confounded with parameters already in the model. Thus, the

researcher must make a decision concerning the relative fishing powers of the gillnets if fitting gamma or log-normal selection curves. The data cannot do it for her.

References

Baranov, F. I. 1948. Theory and assessment of fishing gear. *In* Theory of fishing with gillnets. Chap. 7. Pishchepromizdat, Moscow. (Translation from Russian by Ontario Dept of Lands, Maple, Ont., 45 pp.)

Becker, R. A., Chambers, J. M., and Wilks, A. R. 1988. The new *S* language. Wadsworth and Brooks/Cole. 702 pp.

Borgstrøm, R. 1989. Direct estimation of gill-net selectivity for roach (*Rutilus rutilus* (L.)) in a small lake. Fisheries Research, 7: 289–298.

Boy, V., and Crivelli, A. J. 1988. Simultaneous determination of gillnet selectivity and population age-class distribution for two cyprinids. Fisheries Research, 6: 337–345.

Chambers, J. M., and Hastie, T. J. 1992. Statistical models in *S*. Wadsworth and Brooks/Cole. 608 pp.

Feller, W. 1968. An introduction to probability theory and its applications. Vol. I, 3rd edit. Wiley, New York.

Gulland, J. A. 1983. Fish stock assessment. A manual of basic methods. FAO/Wiley series on food and agriculture, Vol. 1.

Hamley, J. M. 1975. Review of gillnet selectivity. Journal of the Fisheries Research Board of Canada, 32: 1943–1969.

Hamley, J. M., and Regier, H. A. 1973. Direct estimates of gillnet selectivity to walleye (*Stizostedion vitreum vitreum*). Journal of the Fisheries Research Board of Canada, 30: 817–830.

Hastie, T. J., and Tibshirani, R. J. 1990. Generalized additive models. Chapman and Hall, London. 335 pp.

Helser, T. E., Condrey, R. E., and Geaghan, J. P. 1991. A new method of estimating gillnet selectivity, with an example for spotted seatrout, *Cynocion nebulosus*. Canadian Journal of Fisheries and Aquatic Sciences, 48: 487–492.

Helser, T. E., Geaghan, J. P., and Condrey, R. E. 1994. Estimating size composition and associated variances of a fish population from gillnet selectivity, with an example for spotted seatrout *Cynoscion nebulosus*. Fisheries Research, 19: 65–86.

Henderson, B. A., and Wong, J. L. 1991. A method for estimating gillnet selectivity of walleye (*Stizostedion vitreum vitreum*) in multimesh multifilament gill nets in Lake Erie, and its application. Canadian Journal of Fisheries and Aquatic Science, 48: 2420–2428.

Holst, R., and Moth-Poulsen, T. 1995. Numerical recipes and statistical methods for gillnet selectivity. ICES CM 1995B/:18, 22 pp.

Holt, S. J. 1963. A method for determining gear selectivity and its application. ICNAF Special Publication, 5: 106–115.

Kirkwood, G. P., and Walker, T. I. 1986. Gill net mesh selectivities for Gummy Shark, *Mustelus antarcticus* Günther, taken in south-eastern Australian waters. Australian Journal of Marine and Freshwater Research, 37: 689–697.

McCullagh, P., and Nelder, J. A. 1989. Generalized linear models, 2nd edit. Chapman and Hall, London. 511 pp.

Millar, R. B. 1992. Estimating the size-selectivity of fishing gear by conditioning on the total catch. Journal of the American Statistical Association, 87: 962–968.

Millar, R. B., and Walsh, S. J. 1992. Analysis of trawl selectivity studies with an application to trouser trawls. Fisheries Research, 13: 205–220.

Millar, R. B. 1995. The functional form of hook and gillnet selection curves can not be determined from comparative catch data alone. Canadian Journal of Fisheries and Aquatic Science, 52: 883–891.

Myers, R. A., and Hoenig, J. M. In press. Estimates of gear selectivity from multiple tagging experiments. Canadian Journal of Fisheries and Aquatic Science.

Regier, H. A., and Robson, D. S. 1966. Selectivity of gillnets, especially to lake whitefish. Journal of the Fisheries Research Board of Canada, 23: 423–454.

SAS Institute Inc. 1988. SAS/STAT User's guide, release 6.03 Edition. SAS Institute Inc. Cary, NC. 1028 pp.

Wileman, D. A., Ferro, R. S. T., Fonteyne, R., and Millar, R. B. 1996. Manual of methods of measuring the selectivity of towed fishing gears. ICES Cooperative Research Report. No. 215, Copenhagen. 126 pp.

Appendix A

By way of example, if the selection curves are assumed to be normal shaped then

$$r_j(L) = \exp\left(-\frac{(L-\mu_j)^2}{2\sigma_j^2}\right).$$

If geometric similarity is assumed then the spread of gillnet j, σ_j, is proportional to the mesh size, m_j. A convenient parameterization is

$$\mu_j = k_1 \cdot m_j \text{ and } \sigma_j^2 = k_2 \cdot m_j^2, \tag{3}$$

where parameters k_1 and k_2 are to be estimated. Then,

$$\begin{aligned}
\log(v_{Lj}) &= \log(p_j) + \log(\lambda_L) + \log(r_j(L)) \\
&= \log(p_j) + \log(\lambda_L) - \frac{(L-\mu_j)^2}{2\sigma_j^2} \\
&= \log(p_j) + \log(\lambda_L) - \frac{(L-k_1 \cdot m_j)^2}{2k_2 \cdot m_j^2} \\
&= \log(p_j) + \log(\lambda_L) - \frac{k_1^2}{2k_2} + \frac{k_1}{k_2} \cdot \frac{L}{m_j} - \frac{1}{2k_2} \cdot \left(\frac{L}{m_j}\right)^2 \\
&= \log(p_j) + \log(\lambda_L) + \beta_0 + \beta_1 \cdot x_{Lj} + \beta_2 \cdot x_{Lj}^2, \qquad (4)
\end{aligned}$$

where

$$\beta_0 = -\frac{k_1^2}{2k_2} \qquad \beta_1 = \frac{k_1}{k_2} \qquad \beta_2 = -\frac{1}{2k_2}$$

and $x_{Lj} = L/m_j$.

If no form of the population length distribution is assumed then $\log(\lambda_L)$ is fitted as a factor. In doing so, the constant β_0 becomes redundant as it is confounded with the overall level of the $\log(\lambda_L)$ factor. If, in addition, equal relative fishing intensities are assumed then the $\log(p_j)$ terms can be ignored and the model is given by Equation (2) in the text.

Estimates of k_1 and k_2 are derived from the fitted estimates of β_1 and β_2. In this case $\beta_1 = k_1/k_2$

and $\beta_2 = -1/2k_2$, and hence $k_2 = -1/2\beta_2$ and $k = -\beta_1/2\beta_2$.

If it is assumed that the fishing intensities are proportional to mesh size then (to within a constant that can be ignored), $\log(p_j) = \log(m_j)$ and the model becomes

$$\log(v_{Lj}) = \log(m_j) + \text{factor}(L) + \beta_1 \cdot x_{Lj} + \beta_2 \cdot x^2_{Lj}$$

which is fitted by treating $\log(m_j)$ as an offset (Appendix B).

In some cases the log-linear form will also result from models where a population length distribution is assumed. For example, if the population length distribution is assumed to be normal with mean θ and variance τ^2 then

$$\lambda_L = N \exp\left(-\frac{(L-\theta)^2}{2\tau^2}\right).$$

In this case,

$$\begin{aligned}\log(v_{Lj}) &= \left[\log(N) - \frac{k_1^2}{2k_2} - \frac{\theta^2}{2\tau^2}\right] + \left[\frac{\theta}{\tau^2}\right]L + \left[-\frac{1}{2\tau^2}\right]L^2 \\ &\quad + \left[\frac{k_1}{k_2}\right]\frac{L}{m_j} + \left[-\frac{1}{2k_2}\right]\left(\frac{L}{m_j}\right)^2 \\ &= \beta_0 + \beta_1 L + \beta_2 L^2 + \beta_3 x_{Lj} + \beta_4 x^2_{Lj}.\end{aligned}$$

Similar derivations apply for other choices of selection curves (Table 1).

Appendix B

Indirect selectivity data are commonly presented in tabular form. However, both Splus and SAS view each mesh size and length class combination as an observation. Each observation contains three variables – mesh size, length class and catch.

Splus code

In Splus the three variables are represented as three vectors, containing the length class, mesh size, and catch, respectively. The kth elements of the three vectors constitute the data for the kth observation (i.e. the kth mesh size and length class combination.) Thus, for the Fraser River sockeye data (Table 2) the vectors are each of length 88. Here, the three vectors are called `lens`, `msizes` and `catch`, respectively.

Assuming equal fishing intensities of the gillnets, the Splus code for fitting the proportional spread normal shaped selection curves is:

```
var1<-lensmsizes
var2<-var1^2
glm(catch~var1+var2+as.factor(lens),
                          family=poisson)
```

where `glm` is the generalized linear model function. If fishing intensity is proportional to mesh size then

```
glm(catch~var1+var2+as.factor(lens)+
    offset(log(msizes)),family=poisson).
```

The modifications to the log-linear form for incorporation of a normal shaped population length distribution are given in Appendix A. Alternatively, one could fit a model where the population length distribution is a smooth function of length. The log-linear form then no longer applies, and instead one uses the generalized additive model function gam. For example,

```
gam(catch~var1+var2+lo(lens),
                          family=poisson).
```

See Chambers and Hastie (1992) for more detail.

SAS code

In a SAS data step the gillnet dataset, `gilldata`, is created with three variables per observation, `lens`, `msizes`, and `catch`, respectively. The actual SAS data step is left to the reader.

Assuming equal fishing intensities of the gillnets, the SAS code for fitting the proportional spread normal shaped selection curves is:

```
Data;
 set gilldata;
 var1=lens/msizes;
 var2=var1**2;
 oset=log(msizes);

Proc genmod;
 class lens;
 model catch=lens var1 var2/dist=
                                  Poisson;
```

and if fishing intensity is proportional to mesh size then

```
Proc genmod;
class lens;
model catch=lens var1 var2/dist=Poisson
                                offset=oset;
```